3b_pair_information_breakdown

Auto-generated from `2_pipeline/3b_pair_information_breakdown/out/` on 2026-10-13 17:05. Re-run `make_progress_md.py` after the script's output changes -- don't hand-edit this file.

Tables

Pair information breakdown

session_id	animal_id	window	target	roi_name	labeled_grouped	cell_i	cell_j	pair_rank	I_total	I_1	I_2	I_lin	I_ind	I_sigsim
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	AL	unl	LPE10884_2024_01_12_7_0039	LPE10884_2024_01_12_7_0198	2	0.003544	0.01135	0.003618	0.01496	0.02973	0.01476
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	AL	unl	LPE10884_2024_01_12_7_0142	LPE10884_2024_01_12_7_0278	4	0	0.001038	0	0.001038	0	-0.001038
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	AL	unl	LPE10884_2024_01_12_7_0035	LPE10884_2024_01_12_7_0220	1	0	0	0.001472	0.001472	0	-0.001472
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	AL	unl	LPE10884_2024_01_12_7_0116	LPE10884_2024_01_12_7_0136	0	0	0.01514	0	0.01514	0.02107	0.005925
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	AL	unl	LPE10884_2024_01_12_7_0076	LPE10884_2024_01_12_7_0177	3	0	0	0	0	0	0
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	RSP	unl	LPE10884_2024_01_12_6_0002	LPE10884_2024_01_12_6_0053	0	0	0	0	0	0.001888	0.001888
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	RSP	unl	LPE10884_2024_01_12_6_0032	LPE10884_2024_01_12_6_0046	1	0.06252	0	0.04768	0.04768	0.04078	-0.006899
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	RSP	unl	LPE10884_2024_01_12_6_0024	LPE10884_2024_01_12_6_0028	4	0.04089	0	0.0278	0.0278	0.04618	0.01838
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	RSP	unl	LPE10884_2024_01_12_6_0019	LPE10884_2024_01_12_6_0062	3	0.03108	0.01513	0.004434	0.01956	0.01208	-0.007485
LPE10884_2024_01_12	LPE10884	pre_stim	zero_to80_above80	RSP	unl	LPE10884_2024_01_12_6_0012	LPE10884_2024_01_12_6_0025	2	0.01857	0.007489	0.02162	0.02911	0.008077	-0.02103
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	AL	unl	LPE10884_2024_01_12_7_0039	LPE10884_2024_01_12_7_0198	2	0.02597	0.03	0	0.03	0.04165	0.01165
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	AL	unl	LPE10884_2024_01_12_7_0142	LPE10884_2024_01_12_7_0278	4	0	0	0.001471	0.001471	0.002215	0.0007431
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	AL	unl	LPE10884_2024_01_12_7_0035	LPE10884_2024_01_12_7_0220	1	0	0	0.007715	0.007715	0	-0.007715
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	AL	unl	LPE10884_2024_01_12_7_0116	LPE10884_2024_01_12_7_0136	0	0	0.003802	0	0.003802	0.005164	0.001362
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	AL	unl	LPE10884_2024_01_12_7_0076	LPE10884_2024_01_12_7_0177	3	0	0	0	0	0	0
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	RSP	unl	LPE10884_2024_01_12_6_0002	LPE10884_2024_01_12_6_0053	0	0.04312	0	0.004343	0.004343	0.01143	0.007088
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	RSP	unl	LPE10884_2024_01_12_6_0032	LPE10884_2024_01_12_6_0046	1	0.06895	0	0.06434	0.06434	0.05879	-0.00555
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	RSP	unl	LPE10884_2024_01_12_6_0024	LPE10884_2024_01_12_6_0028	4	0.01334	0	0.02988	0.02988	0.02376	-0.006125
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	RSP	unl	LPE10884_2024_01_12_6_0019	LPE10884_2024_01_12_6_0062	3	0.01961	0	0.01756	0.01756	0	-0.01756
LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	RSP	unl	LPE10884_2024_01_12_6_0012	LPE10884_2024_01_12_6_0025	2	0	0.009321	0.002105	0.01143	0.01314	0.001712
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	AL	unl	LPE10884_2024_01_12_7_0039	LPE10884_2024_01_12_7_0198	2	0	0	0.004516	0.004516	0	-0.004516
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	AL	unl	LPE10884_2024_01_12_7_0142	LPE10884_2024_01_12_7_0278	4	0	0	0	0	0.004833	0.004833
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	AL	unl	LPE10884_2024_01_12_7_0035	LPE10884_2024_01_12_7_0220	1	0.01346	0	0	0	0	0
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	AL	unl	LPE10884_2024_01_12_7_0116	LPE10884_2024_01_12_7_0136	0	0	0.012	0	0.012	0.005619	-0.006386
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	AL	unl	LPE10884_2024_01_12_7_0076	LPE10884_2024_01_12_7_0177	3	0.01055	0	0	0	0	0
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	RSP	unl	LPE10884_2024_01_12_6_0002	LPE10884_2024_01_12_6_0053	0	0.0217	0	0	0	0.006059	0.006059
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	RSP	unl	LPE10884_2024_01_12_6_0032	LPE10884_2024_01_12_6_0046	1	0	0	0	0	0	0
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	RSP	unl	LPE10884_2024_01_12_6_0024	LPE10884_2024_01_12_6_0028	4	0.01216	0.007227	0	0.007227	0	-0.007227
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	RSP	unl	LPE10884_2024_01_12_6_0019	LPE10884_2024_01_12_6_0062	3	0.02646	0.01003	0	0.01003	0	-0.01003
LPE10884_2024_01_12	LPE10884	pre_stim	zero_vs_nonzero	RSP	unl	LPE10884_2024_01_12_6_0012	LPE10884_2024_01_12_6_0025	2	0.007711	0	0	0	0.001439	0.001439

(showing first 15 of 25 columns; showing first 30 of 7248 rows -- see the linked CSV for the rest)

pair_information_breakdown.csv

Pair information breakdown by group session

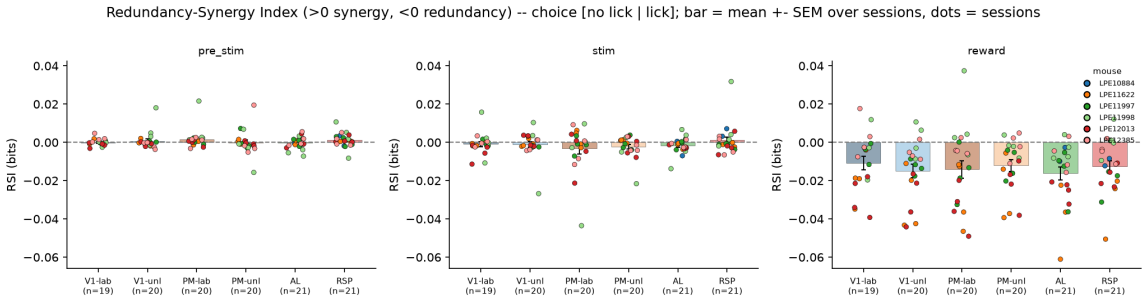
group	unit	animal_id	window	target	n_pairs	I_total	I_lin	I_sigsim	I_cor	RSI	I_corind_approx	I_cordep_approx	frac_synergistic
AL_unl	LPE10884_2024_01_12	LPE10884	pre_stim	choice	5	0.001126	0.00163	0.001173	-0.001677	-0.0005037	0.04753	-0.04921	0.2
AL_unl	LPE10884_2024_01_17	LPE10884	pre_stim	choice	5	0.0173	0.01559	-0.0002086	0.001918	0.00171	0.07606	-0.07414	0.4
AL_unl	LPE11622_2024_02_22	LPE11622	pre_stim	choice	5	0.0005034	0.001672	-0.001117	-5.211e-05	-0.001169	0.2608	-0.2609	0.2
AL_unl	LPE11622_2024_02_23	LPE11622	pre_stim	choice	5	0.003917	0.002967	0.0004766	0.000473	0.0009496	0.2523	-0.2518	0.2
AL_unl	LPE11622_2024_02_26	LPE11622	pre_stim	choice	5	0.000699	0.002038	0.0004218	-0.001761	-0.001339	0.318	-0.3198	0.2
AL_unl	LPE11997_2024_04_10	LPE11997	pre_stim	choice	5	0.0026	0.001023	-0.0005606	0.002137	0.001576	0.07699	-0.07486	0.6
AL_unl	LPE11997_2024_04_16	LPE11997	pre_stim	choice	5	0.001867	0.003963	-0.001528	-0.0005683	-0.002096	0.1785	-0.1791	0.2
AL_unl	LPE11997_2024_04_17	LPE11997	pre_stim	choice	5	0.00235	0.001299	0.0001927	0.0008577	0.00105	0.258	-0.2571	0.4
AL_unl	LPE11997_2024_04_18	LPE11997	pre_stim	choice	5	0.007536	0.008555	-0.003639	0.00262	-0.001019	0.1398	-0.1372	0.4
AL_unl	LPE11998_2024_04_24	LPE11998	pre_stim	choice	5	0.03375	0.03921	-0.007742	0.002289	-0.005453	0.102	-0.0997	0.4
AL_unl	LPE11998_2024_04_25	LPE11998	pre_stim	choice	5	0.009509	0.006198	-0.003769	0.00708	0.003311	0.03371	-0.02663	0.4
AL_unl	LPE11998_2024_04_29	LPE11998	pre_stim	choice	5	0.007121	0.01097	0.001877	-0.00573	-0.003852	0.06179	-0.06752	0.2
AL_unl	LPE11998_2024_05_01	LPE11998	pre_stim	choice	5	0.002004	0.009203	-0.005282	-0.001917	-0.007199	0.01719	-0.01911	0
AL_unl	LPE12013_2024_04_25	LPE12013	pre_stim	choice	5	0.00112	0.003981	-0.001152	-0.001709	-0.002861	0.1025	-0.1042	0.2
AL_unl	LPE12013_2024_04_26	LPE12013	pre_stim	choice	5	0.008065	0.003543	0.00223	0.002292	0.004523	0.1381	-0.1358	0.8
AL_unl	LPE12013_2024_04_29	LPE12013	pre_stim	choice	5	0	0.000819	-0.0001366	-0.0006824	-0.000819	0.107	-0.1077	0
AL_unl	LPE12013_2024_04_30	LPE12013	pre_stim	choice	5	0.002838	0.000962	0.0003393	0.001536	0.001876	0.1623	-0.1608	0.4
AL_unl	LPE12385_2024_06_15	LPE12385	pre_stim	choice	5	0	0.001537	0.001256	-0.002792	-0.001537	0.0266	-0.02939	0
AL_unl	LPE12385_2024_06_16	LPE12385	pre_stim	choice	5	0.009894	0.008015	-0.004387	0.006266	0.001879	0.02523	-0.01896	0.6
AL_unl	LPE12385_2024_06_17	LPE12385	pre_stim	choice	5	0.00891	0.003311	0.0008582	0.004741	0.005599	0.03989	-0.03515	0.4
AL_unl	LPE12385_2024_06_27	LPE12385	pre_stim	choice	5	0.01207	0.0128	-0.002086	0.001359	-0.0007262	0.02545	-0.02409	0.4
AL_unl	LPE10884_2024_01_12	LPE10884	pre_stim	pct_0-20_20-80_80-100	5	0.005194	0.008597	0.001208	-0.004612	-0.003404	0.1098	-0.1144	0
AL_unl	LPE10884_2024_01_17	LPE10884	pre_stim	pct_0-20_20-80_80-100	5	0.001185	0.003793	-0.002412	-0.0001956	-0.002608	0.1414	-0.1416	0.4
AL_unl	LPE11622_2024_02_22	LPE11622	pre_stim	pct_0-20_20-80_80-100	5	0.0005844	0.001393	-2.501e-05	-0.0007834	-0.0008084	0.2683	-0.2691	0.2
AL_unl	LPE11622_2024_02_23	LPE11622	pre_stim	pct_0-20_20-80_80-100	5	3.947e-06	0.001778	-0.001673	-0.0001008	-0.001774	0.3717	-0.3718	0
AL_unl	LPE11622_2024_02_26	LPE11622	pre_stim	pct_0-20_20-80_80-100	5	0.0007661	0.001656	-0.0006624	-0.0002279	-0.0008903	0.388	-0.3883	0.2
AL_unl	LPE11997_2024_04_10	LPE11997	pre_stim	pct_0-20_20-80_80-100	5	0.001901	0.004628	-0.002775	4.898e-05	-0.002726	0.1296	-0.1295	0.2
AL_unl	LPE11997_2024_04_16	LPE11997	pre_stim	pct_0-20_20-80_80-100	5	0.01042	0.008057	-0.002897	0.005255	0.002358	0.1706	-0.1654	0.2
AL_unl	LPE11997_2024_04_17	LPE11997	pre_stim	pct_0-20_20-80_80-100	5	0.002716	0.00255	0.0002122	-4.666e-05	0.0001656	0.1532	-0.1533	0.4
AL_unl	LPE11997_2024_04_18	LPE11997	pre_stim	pct_0-20_20-80_80-100	5	0.003301	0.001882	0.0006739	0.0007445	0.001418	0.1879	-0.1871	0.6

(showing first 30 of 1464 rows -- see the linked CSV for the rest)

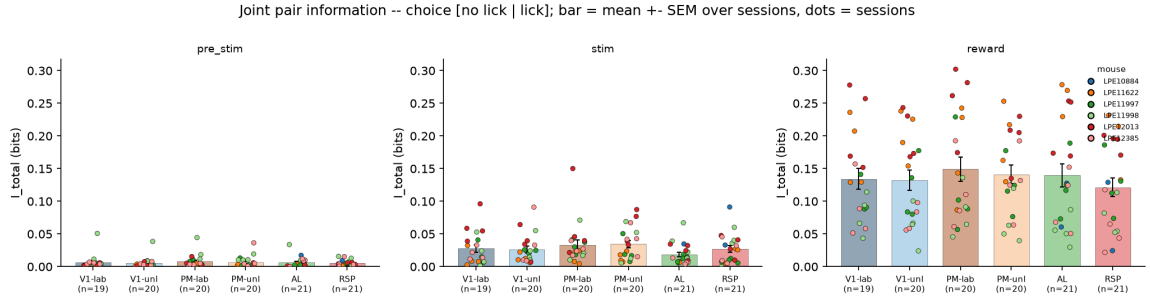
pair_information_breakdown_by_group__session.csv

Figures

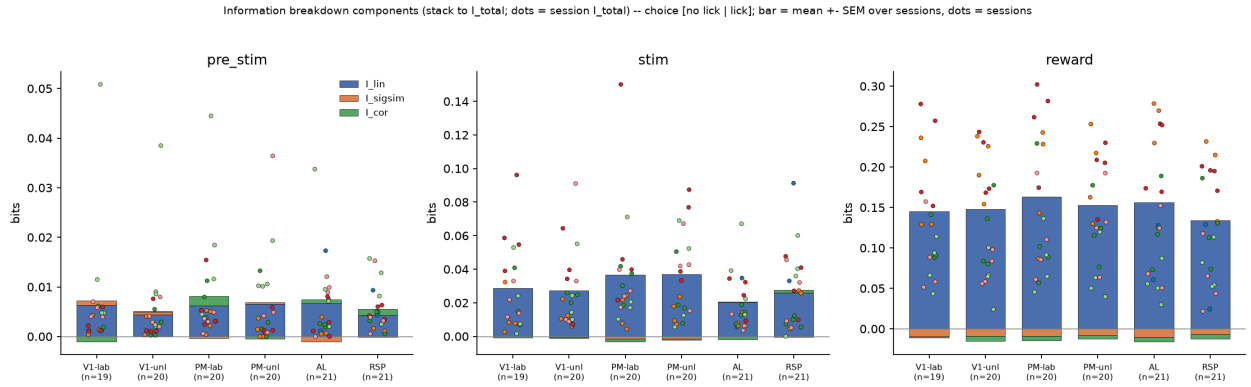
Rsi by group



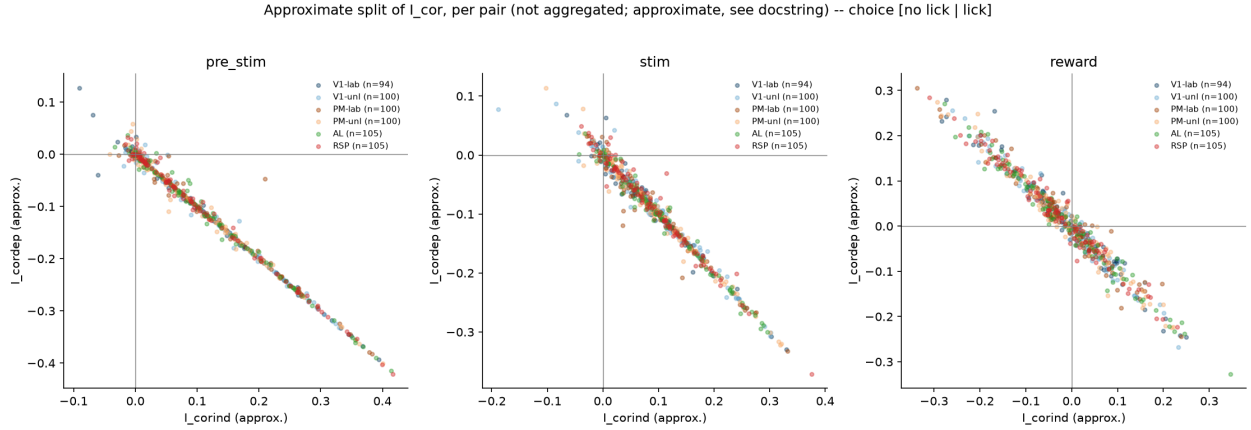
1b itotal by group



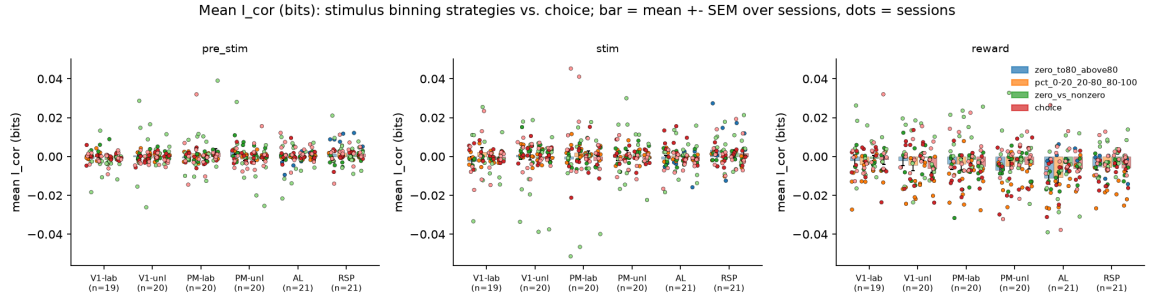
Component breakdown



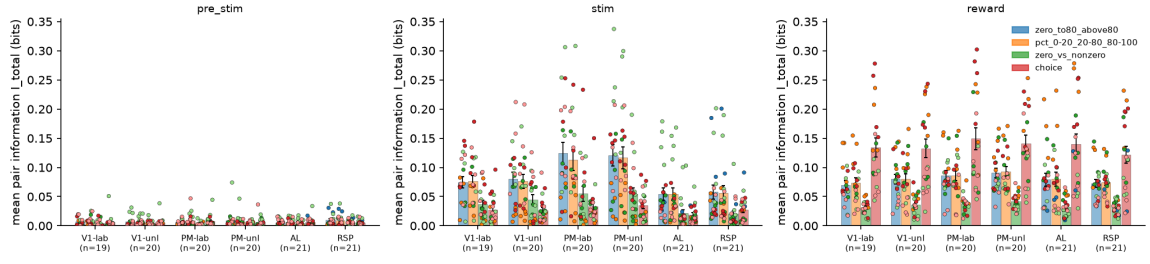
Corind vs cordep approx



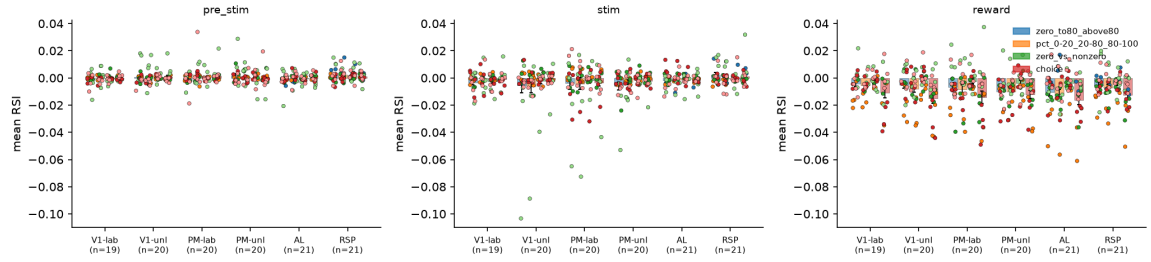
Compare targets I cor



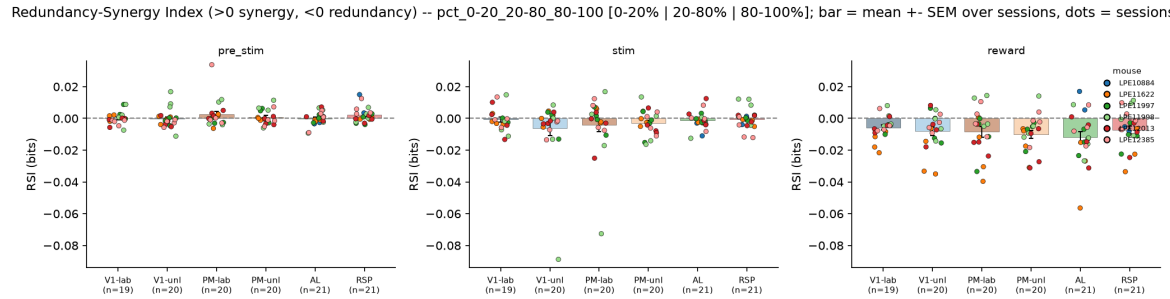
Compare targets I total



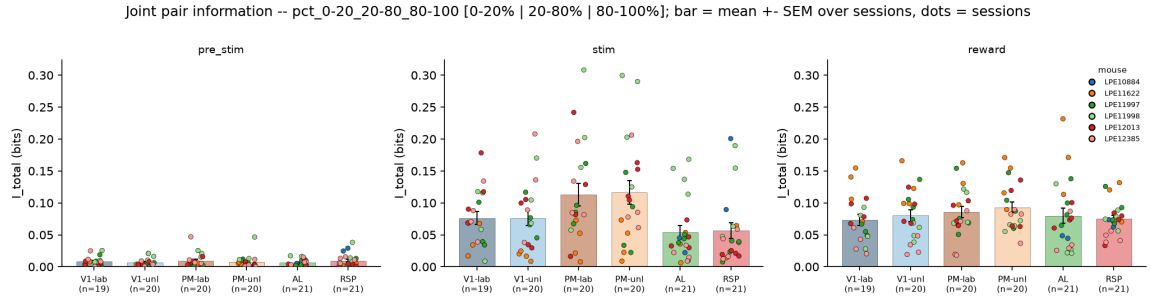
Compare targets RSI



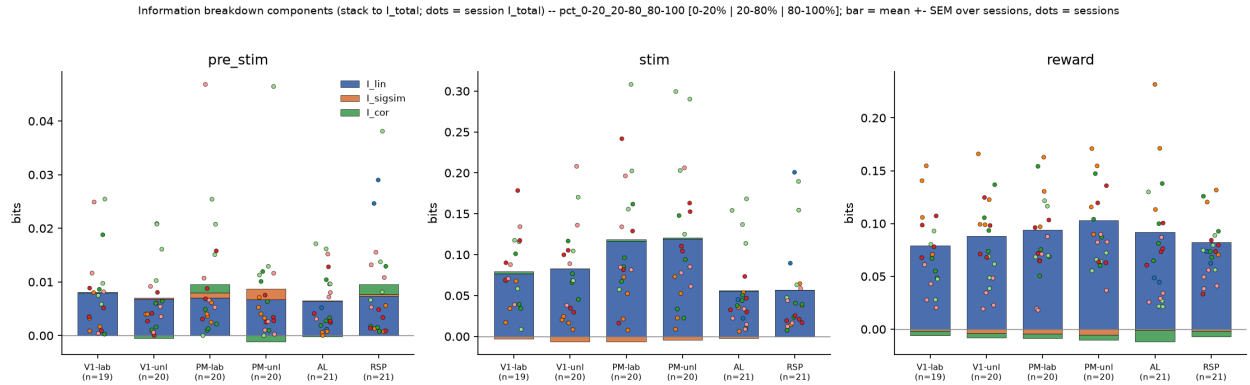
Rsi by group



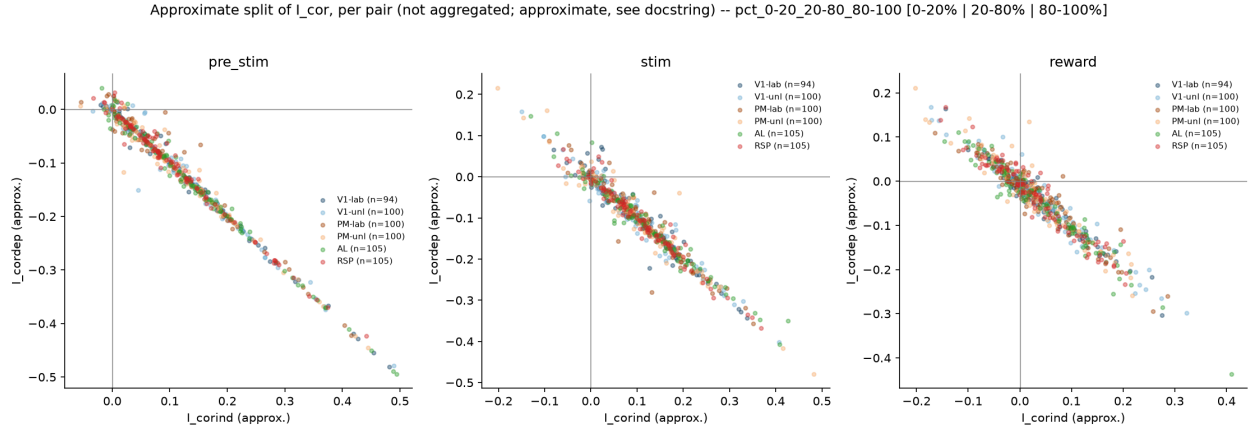
1b itotal by group



Component breakdown

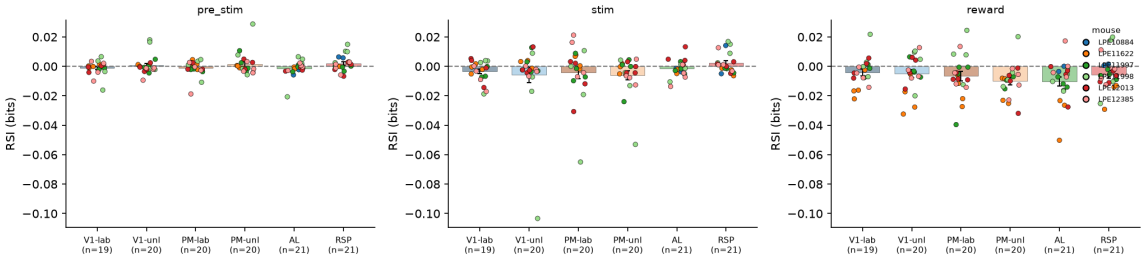


Corind vs cordep approx



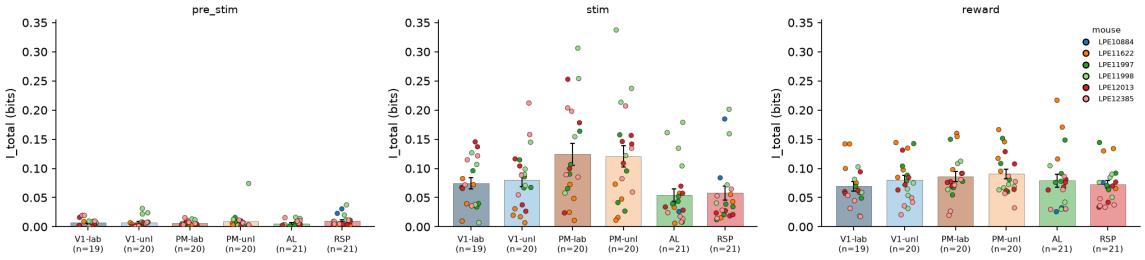
Rsi by group

Redundancy-Synergy Index (>0 synergy, <0 redundancy) -- zero_to80_above80 [0 | <=80% | >80%]; bar = mean +- SEM over sessions, dots = sessions



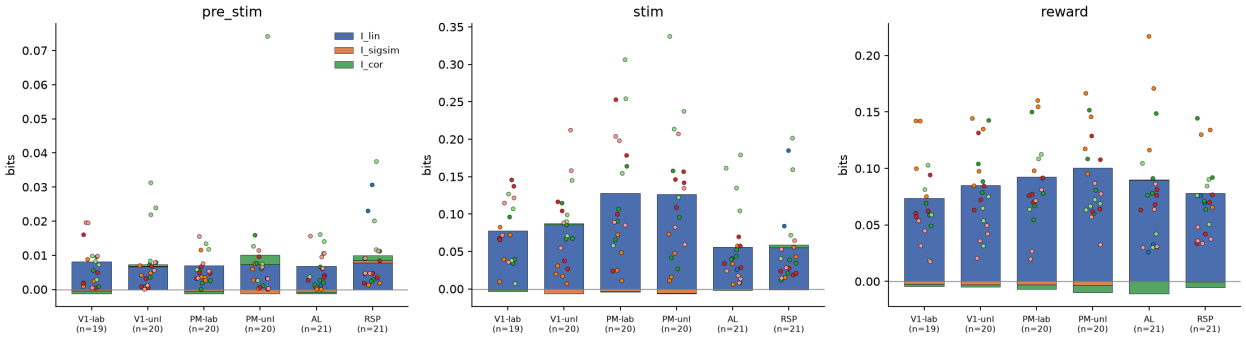
1b itotal by group

Joint pair information -- zero_to80_above80 [0 | <=80% | >80%]; bar = mean +- SEM over sessions, dots = sessions



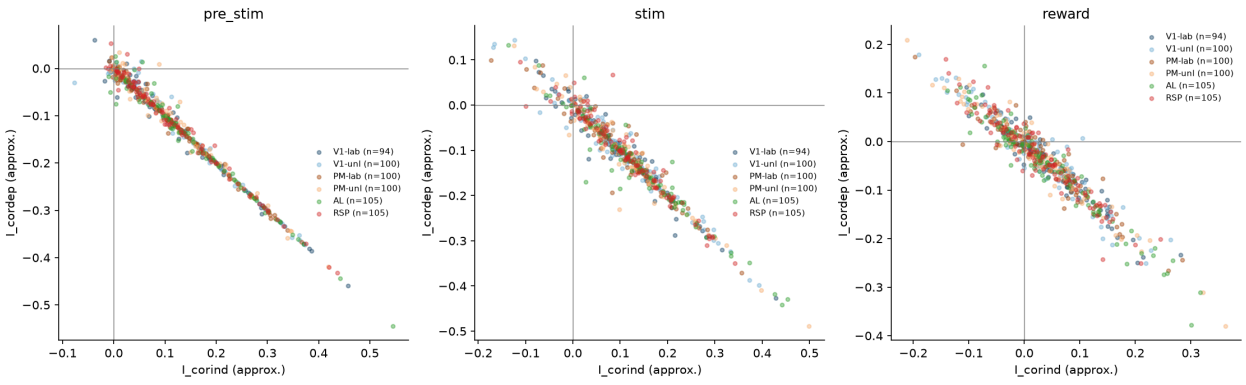
Component breakdown

Information breakdown components (stack to I_total; dots = session I_total) -- zero_to80_above80 [0 | <=80% | >80%]; bar = mean +- SEM over sessions, dots = sessions



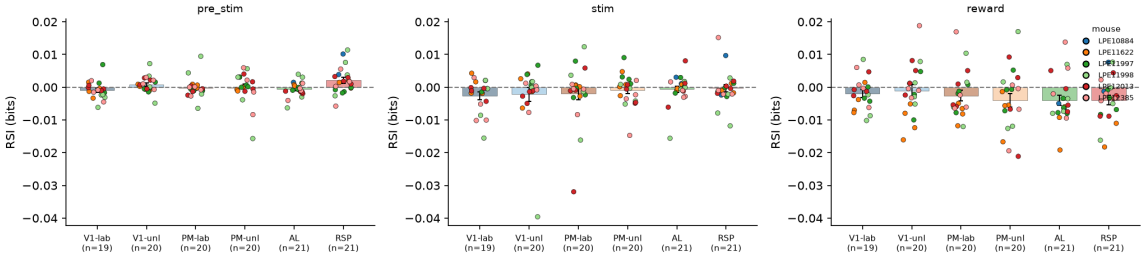
Corind vs cordep approx

Approximate split of I_cor, per pair (not aggregated; approximate, see docstring) -- zero_to80_above80 [0 | <=80% | >80%]

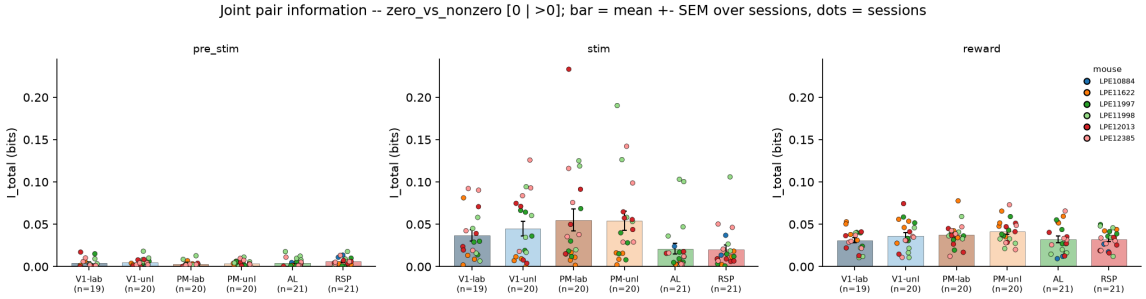


Rsi by group

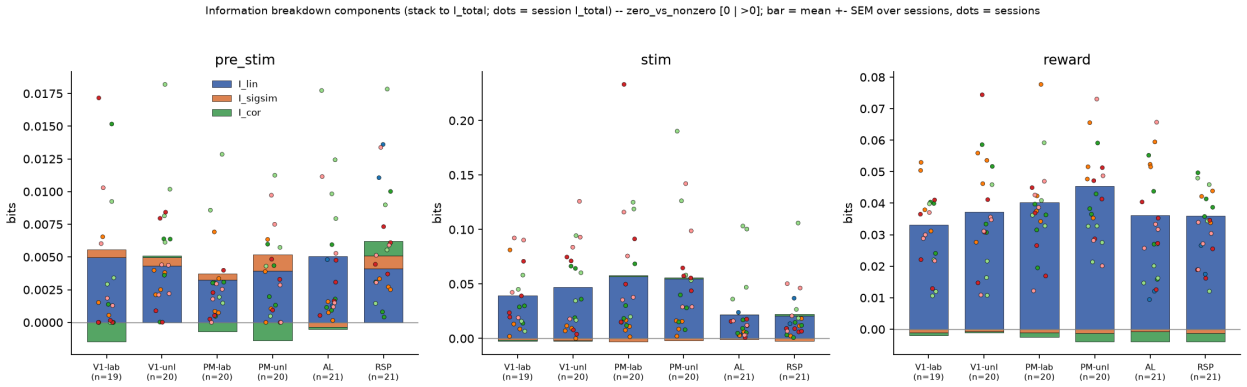
Redundancy-Synergy Index (>0 synergy, <0 redundancy) -- zero_vs_nonzero [0 | >0]; bar = mean +- SEM over sessions, dots = sessions



1b itotal by group



Component breakdown



Corind vs cordep approx

